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The Role of Statistical Learning in Attentional Guidance During Search Through Naturalistic Scenes

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Regularities within the environment have been shown to guide attention and facilitate efficient search; however, previous work utilized real-world objects that participants were already familiar with, making it difficult to discern the relative contribution of statistical learning and semantic knowledge to this effect. In Experiment 1, we associated novel objects (with no apparent semantic meaning) with regions in kitchen scenes (floor, countertop, wall) but placed them in the associated region of the scene on 75% of trials; we found that participants were faster on trials when the target appeared in its high-probability location, consistent with learning-dependent guidance. In Experiment 2, we looked to see if participants could learn multiple context-dependent object–location associations. We found that when the associated location of an object varied by the context provided by the category of scene, participants were unable to learn such associations. In Experiment 3, further showed that the learning in one context containing object–location associations similarly guided attention in a second context that did not contain the same contingency. Overall, our results suggest that learned object–location associations can guide attention in the absence of semantic knowledge concerning the objects, suggesting a role for statistical learning in scene-based guidance, but such learning does not present with the context specificity associated with semantic guidance.

Keywords: statistical learning, visual search, scene grammar, naturalistic scenes

Imagine yourself tasked with searching for a television remote. With various brands of remotes available, all with distinct qualities and features, finding a remote in a complex environment might seem in some ways challenging. However, despite differences in appearance, remotes tend to be found in consistent locations like a side table near a couch or a nightstand in a bedroom. By utilizing this regularity, we do not have to check every possible position in a given scene and can, instead, begin search by first looking for an object in its most probable locations. This ability to encode and maintain regularities of where an object occurs within the environment is critical for the ability to efficiently search through scenes. The rules and regularities that guide search for objects in scenes have been referred to as *scene grammar* (Võ et al., 2019).

Scene grammar reflects the knowledge someone gains through experiencing the configuration of objects within the environment. Objects are generally not positioned randomly within environments but rather occur with regularities based on function and context. For example, a coffee pot is most likely to be found on a table, and light switches are more likely to be found on the wall near a door or entryway. These regularities can be learned to create expectations about where an object may occur in the scene to assist and guide visual search (Bar, 2004; Draschkow & Võ, 2017).

The rules governing the makeup of a scene have been divided into two distinct categories: semantic and syntactic scene knowledge

(Võ & Henderson, 2009). Semantic knowledge refers to knowledge about an object and its context (e.g., toilet article belongs in the bathroom), whereas syntactic knowledge refers to knowledge about the physical constraints of an object (i.e., an object rests on a surface due to gravity and dish soap tends to be found next to a sink). A violation of this knowledge has been shown to impair identification and perception of an object (Biederman et al., 1982).

The overall meaning of a scene can be extracted very rapidly (<50 ms); this rapidly extracted scene information is referred to as scene gist (Castelhano & Henderson, 2008; Friedman, 1979; Greene & Oliva, 2009a, 2009b; Potter, 1975, 1976; Rousselet et al., 2005). The gist of a scene must first be extracted in order to utilize semantic information during visual search. Once the scene gist is established, it supplies the semantic context necessary to plan eye movements and guide attention to likely locations of target objects (Castelhano & Henderson, 2007). With the semantic structure of the scene in place, prior knowledge about the target object (including its typical size, shape, and likely location within certain environments) can further guide visual search by narrowing down the spatial regions most likely to contain the object (Malcolm & Henderson, 2010; Torralba et al., 2006).

Other types of memory also play key roles in visual search (for a review, see Le-Hoa Võ & Wolfe, 2015). Memory for specific object locations from previous searches can influence how attention is

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administration, supervision, and writing—review and editing and an equal role in conceptualization and methodology.

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deployed in future searches. For example, contextual cueing demonstrates how repeated exposure to consistent spatial configurations can facilitate search (Chun & Jiang, 1998). Moreover, inhibition of return is a mechanism by which attention is less likely to revisit recently attended locations, promoting exploration of new areas and improving search efficiency over time (Klein, 2000; MacInnes & Klein, 2003; Posner & Cohen, 1984; Z. Wang & Klein, 2010).

In real-world scenes, the context in which a target is presented plays a large role in how attention is deployed. Brockmole et al. (2006) showed that repeated search for an object in a scene was faster when global context remained the same but local context varied than vice versa, demonstrating that in search through naturalistic scenes, global contexts are more heavily weighted than local context when learning object–location associations. Additionally, semantic expectations are critical in guiding attention in real-world scenes. Brockmole and Le-Hoa Võ (2010) found that the relationship between a target (e.g., the letter “T”) and an object (e.g., a pillow) could be used to guide attention and speed up search. When the pillow appeared in a semantically consistent context (i.e., a bedroom or living room), search was facilitated. In contrast, when the pillow was placed in a semantically inconsistent context (i.e., a kitchen), the relationship between the pillow and the target did not aid search performance. This suggests that semantic consistency between objects and their surrounding environment plays a key role in shaping attentional guidance, enabling more efficient search behavior in familiar and meaningful contexts. Other work has utilized scene context to demonstrate that semantically inconsistent objects capture attention automatically and produce longer fixation durations (Henderson et al., 1999; Loftus & Mackworth, 1978; Võ & Henderson, 2009). Together with Brockmole and Le-Hoa Võ, this work showcases that context is a necessary feature for semantic knowledge-based guidance of attention.

While semantic and syntactic knowledge contribute to the facilitation of attention, attentional guidance has also been shown to be shaped by mechanisms that do not rely on explicit knowledge, such as statistical learning, the ability to implicitly extract regularities from the environment through repeated exposure (Clement & Anderson, 2024; Fiser & Aslin, 2001, 2002; Turk-Browne et al., 2005). Statistical learning has been shown to guide attention across different levels of representation including simple visual features (e.g., color or orientation; Chun & Jiang, 1998; Turk-Browne et al., 2005), specific object identities (e.g., feature–category associations; Bahle et al., 2021), or spatial structures of environments (e.g., contextual cueing; Chun & Jiang, 1998). Object features can guide attention to stimuli when they reliably characterize task-relevant objects or categories (Bahle et al., 2021; Clement & Anderson, 2024; Kershner & Hollingworth, 2022). Although statistical learning was initially thought to produce relatively inflexible and persistent attentional biases, recent evidence suggests that these learned regularities can be context-dependent. For example, Zhang and Carlisle (2023) showed that distinct spatial priority maps for different target identities could be flexibly activated according to current task goals. This suggests that statistical learning interacts with contextual information to guide attention in a goal-specific manner.

Previous work demonstrates that scene semantics plays a central role in the guidance of attention in real-world scenes; however, this work has been conducted using familiar real-world objects and corresponding scene grammar. Using real-world objects in this manner carries the limitation that attentional guidance to these

objects was formed throughout the lifespan of an individual through experience; therefore, it is difficult to discern what factors are critical in the development of such learning-dependent attentional guidance. Two possible accounts could be proposed for how this learning occurs: a *semantic knowledge* account and a *statistical learning* account. According to the semantic knowledge account, attentional guidance for a given object is predicated on the ability to leverage declarative knowledge concerning the function of the object and its relationship with other objects and the layout of the scene. By the statistical learning account, facilitation to a target item could rather be explained through a reliance on the frequency with which a particular object has been paired with objects and regions within previously searched environments. We can test the statistical learning account by utilizing novel complex geometric objects that lack semantic meaning; such objects do not possess a meaningful relationship with other objects and regions within scenes based on their (lack of) function, but they are visually distinct such that the learning of specific object–location relationships should be possible (Fiser & Aslin, 2001, 2002; Turk-Browne et al., 2005).

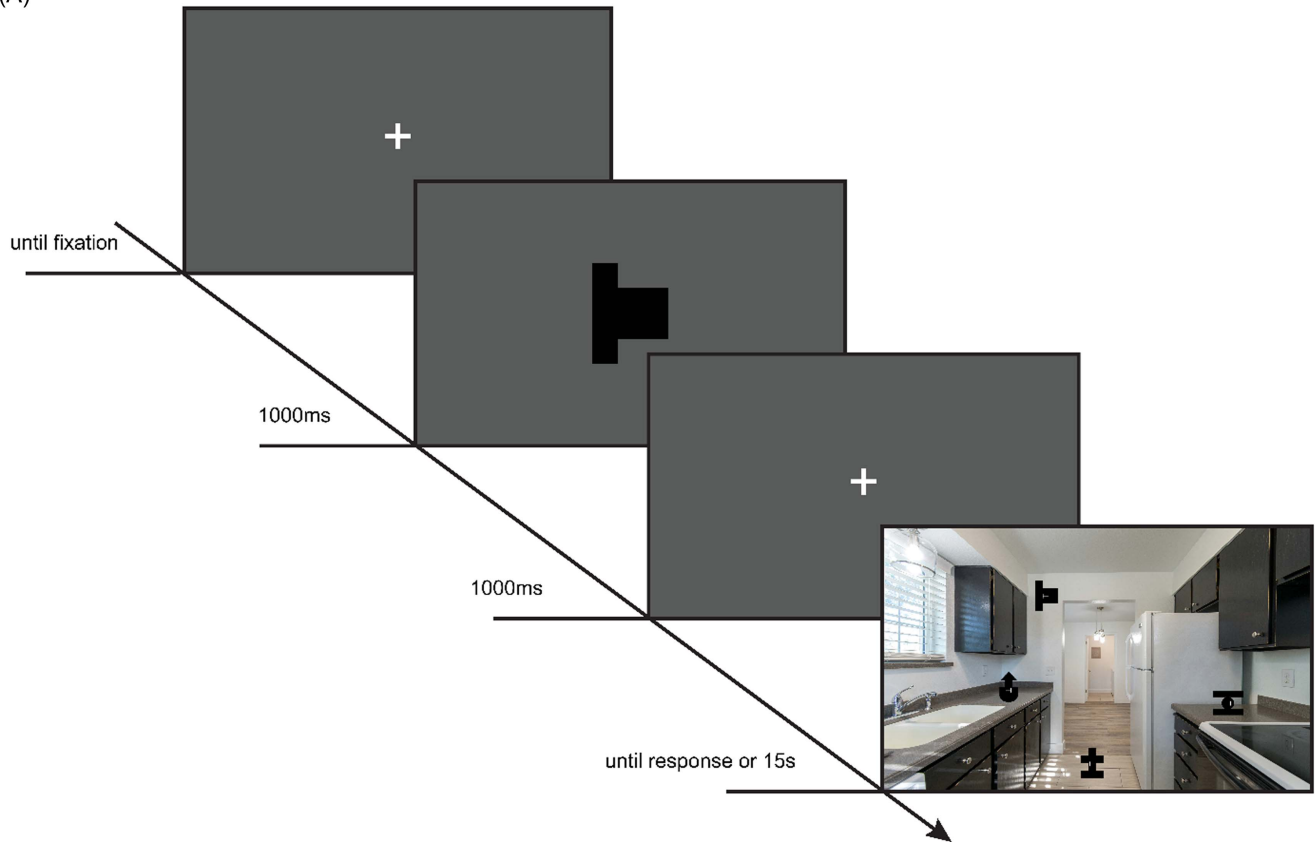
The effects of semantics and statistical learning on the guidance of attention are evident in the efficiency of search as measured from RT in identifying and reporting a target (Anderson, 2026; Belke et al., 2008; B. Wang & Theeuwes, 2018) as well as the frequency with which eye movements are directed to targets and distractors (Hayes & Henderson, 2019; Hwang et al., 2011; Y. V. Jiang et al., 2014; B. Wang et al., 2019). In the present study, through measuring eye position and behavioral responses, we examined whether participants could learn to exploit statistical regularities between objects and locations in naturalistic scenes that could only be learned through the experience of repeated pairings between the two. To test this, we removed semantic information from target objects by utilizing complex novel shapes with little to no inherent semantic information. This is not to suggest that semantic information does not play a role in guiding attention in our study, as the scenes themselves still contain semantic structure, but the relationship between the target objects and this semantic structure can only be learned from experience given the arbitrary nature of the target objects. Across three experiments, we measured search efficiency (probability of looking at the target on first fixation and response time [RT]), when novel objects were presented in a statistically likely location within a scene (floor, countertop, wall) versus an unlikely location, with different target objects associated with different locations. Experiment 1 sought to establish whether participants are able leverage relationships between objects and locations within scenes of a single category (kitchens), whereas Experiments 2 and 3 probed the scope of such learning, and specifically whether the category of scene (which itself carries semantic information) bounds such learning.

Experiment 1

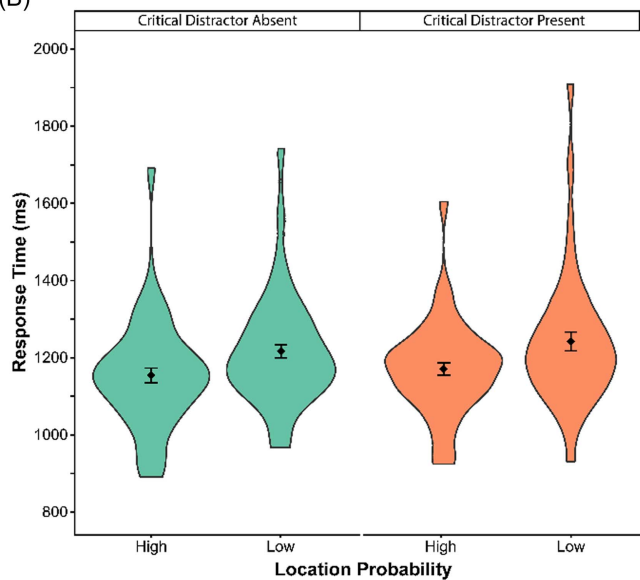
In Experiment 1, we examined whether object–location associations could be learned through statistical relationships between objects and locations within a scene, in the absence of semantic relationships. While eye position was tracked, on each trial, participants were instructed to search for a target object in a subsequent scene as shown in Figure 1A. Each target was probabilistically associated with a specific scene location (wall, countertop, or floor), appearing in that location on 75% of trials. In addition to the target shape, three other distractor shapes appeared in each image,

Figure 1*Sequence of Trial Events and Behavioral Results for Experiment 1*

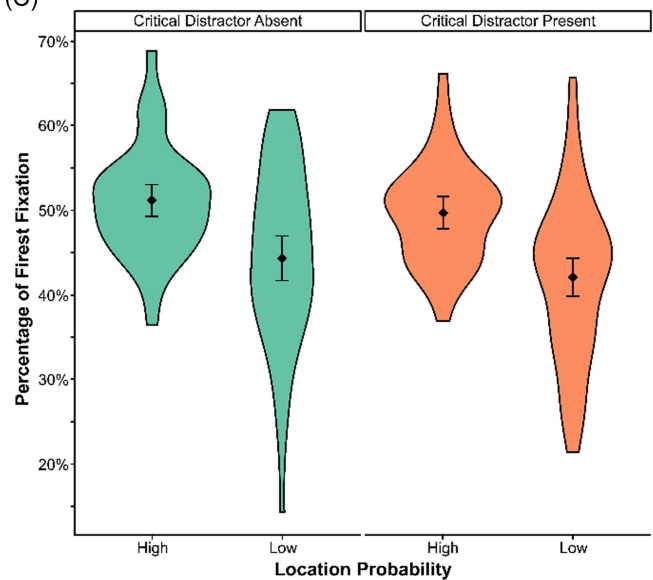
(A)



(B)



(C)



Note. (A) Time course of a trial in Experiment 1. Participants fixated a central cross, viewed a target cue for 1,000 ms (indicating the shape to be searched for in the upcoming display) and then fixated a second cross for 1,000 ms. The search display remained visible until a response was made or 15 s elapsed. After each response, participants received 500-ms feedback indicating accuracy. (B) Response time across location type and critical distractor presence in Experiment 1. Violin plots show the distribution of fixation proportions in each condition. Dots represent condition means, and error bars denote 95% Cousineau–Morey confidence intervals. (C) Percentage of first fixations on the target across location type and critical distractor presence in Experiment 1, shown with the same plotting conventions as in (B). See the online article for the color version of this figure.

positioned in the same categories of location. If participants are faster to find a target when it appears in its high-probability location, it would suggest that participants are able to form object–location associations in the absence of semantic relationships linking them. To additionally speak to whether any observed biases reflect a mere familiarity benefit, we additionally included trials in which a noncued target shape was used as a distractor (i.e., a distractor that could be used for a target on other trials), which required participants to search for the specific cued target shape; we hypothesized that although such noncued target distractors may interfere with search, such interference would not disrupt any effect of the target–location association by adding another familiar, prioritized stimulus.

Method

Participants

A priori power analysis was conducted using G*Power 3.1 (Faul et al., 2007) for a 2×2 repeated-measures ANOVA. Assuming a medium effect size ($f = .25$) and a moderate correlation among repeated measures ($\rho = .5$), the analysis indicated that a minimum of 24 participants would be sufficient to detect at least one simple effect with 80% power ($\alpha = .05$). However, to improve the precision of effect size estimates and help provide sufficient power to detect possible interaction effects and/or smaller effect sizes without the contribution of semantic relationships, we recruited 41 participants from the Texas A&M University community. Participants received course credit as compensation. The sample included 16 males, 28 females, and two individuals who chose not to self-report gender. Participants had an average age of 18.8 ($SD = .93$) and were all between the ages of 18 and 21. One participant was excluded due to low accuracy ($<80\%$). All participants reported normal or corrected-to-normal visual acuity and normal color vision. All participants provided written informed consent, and all study procedures were approved by the Texas A&M University Institutional Review Board.

Apparatus

A Dell OptiPlex 7,040 equipped with MATLAB software and Psychophysics Toolbox extensions (Brainard, 1997) was used to present the stimuli on a Dell P2717H monitor. Responses were entered using a standard U.S.-layout keyboard. The participants viewed the monitor from a distance of approximately 70 cm in a dimly lit room. Eye position was monitored using an Eyelink 1,000-plus desktop mount eyetracker. Head position was maintained using an adjustable chin and forehead rest (SR research).

Stimuli

A total of 96 kitchen scenes were used in the main task and a different set of 24 kitchen scenes were used in the practice block. Scenes were selected via Internet search from online stock images, the change blindness data set (Sareen et al., 2016), and Adobe Stock Images and were selected to contain prominent wall, countertop, and floor locations. Scene stimuli filled the expanse of the monitor, measuring $30.8^\circ H \times 52.2^\circ W$ of visual angle. Targets and distractors were shapes adapted from Fiser and Aslin (2001) and Turk-Browne et al. (2005) and consisted of 22 novel shapes. These shapes all were complex geometric shapes created through combination of simple shapes to ensure that stimuli had no prior associations. Each target

cue measured $7^\circ \times 7^\circ$. When presented within a scene, all shapes measured $3.1^\circ \times 3.1^\circ$ and were rendered black, each with a small white T rotated either 90° left or right embedded in the center. For each participant, four shapes from this set were randomly selected to serve as targets (and critical distractors on trials in which they were not used as the target), while the remaining shapes were used as noncritical distractors. On each trial, one target and three distractors were placed in each scene. Each target shape was given an association with either the wall, countertop, or floor of the scenes; one of each location type was paired with a target, and a repeated location type was paired with the remaining target (randomly selected from the three).¹ For each scene, four locations for the targets and distractors to appear were selected based on where the location of the shape (wall, countertop, or floor) appeared minimally ambiguous and where the black shapes could be easily distinguishable from the background; one location was selected for the wall, countertop, and floor, with a second location for one of the three.

Due to variations in camera angles and scene layout, the exact pixel coordinates for each location type could vary with each scene. In order to control for position effects that might covary with scene locations (e.g., floor locations tend to appear toward the bottom of images, countertops toward the middle, and wall locations toward the top), we leveraged this variation to make most positions have overlap with another type of position in a different scene (e.g., the same pixel location on the screen could be a floor location in one scene and a countertop location in another). On half of all trials, one of the shapes from the set used for targets appeared as a critical distractor (i.e., on trials in which a different target shape was cued) to assess how participants were utilizing the cue.

Design

The experiment consisted of one practice block of 24 trials and six test blocks of 60 trials for a total of 384 trials (including practice). The practice block consisted of six trials for each possible target in which the target appeared in its associated high-probability location on 100% of trials. The test blocks consisted of 96 scenes that could be repeated up to four times across the task. During the test blocks, targets appeared in their associated locations on 75% of trials and across the other two nonassociated locations on 25% of trials. Exactly half of each of high-probability location and low-probability location trials included a critical distractor, the location of which was randomly selected from among the three locations not occupied by the target. Each target was cued on a fourth of all trials. Trial order

¹ To determine whether having two target shapes share the same high-probability location impacted our results, we ran a three-way ANOVA on RT with the same factors as the primary ANOVA (location probability and critical distractor presence) and an additional factor for the duplication of associated location (whether or not a target shared its high-probability location with another target). We found that the main effects of location probability, $F(1, 39) = 28.13, p < .001, \eta_p^2 = .42$, and the presence of the critical distractor remained significant, $F(1, 39) = 6.15, p = .018, \eta_p^2 = .14$. We found no main effect of duplicate location type, $F(1, 39) = .23, p = .64, \eta_p^2 < .01$, however. Nor were any significant interactions observed, including the interactions between location probability and critical distractor presence, $F(1, 39) = .36, p = .55, \eta_p^2 < .01$, location probability and duplication, $F(1, 39) = 1.99, p = .17, \eta_p^2 = .05$, and critical distractor presence and duplication, $F(1, 39) = 3.03, p = .09, \eta_p^2 = .07$, as well as the three-way interaction, $F(1, 39) = 2.83, p = .10, \eta_p^2 = .07$. These results suggest that the primary findings were not a byproduct of participants prioritizing the location associated with two targets regardless of target identity.

was randomized across the experiment with the constraints that the same scene could not appear twice within the same block, no more than two consecutive trials could involve any target in a low-probability location, and a given target could not be cued on more than two consecutive trials. On each trial, two of the shapes were presented with a T rotated 90° left, and two of the shapes were presented with a T rotated 90° right. On half of all trials, the target appeared with its embedded T rotated 90° left and the other half of trials it appeared with its embedded T rotated 90° right.

Procedure

After giving informed consent, participants completed demographics questions, were calibrated on the eyetracker (to confirm trackability), and were then given task instructions. Before each block of trials, eye position was calibrated, and during the task, eye position was manually drift corrected by the experimenter as necessary to acquire central fixation before the next trial occurred. Calibration consisted of a standard 9-point calibration procedure commonly utilized in studies involving scene viewing (e.g., Anderson & Kim, 2018; Liao & Anderson, 2020). Participants were first presented with a fixation display that consisted of a white plus sign presented in the center of the screen on a gray background. The fixation display remained on screen until eye position was registered within 1.4° of the central fixation cross for a duration of 500 ms, being manually drift corrected by the experimenter as necessary to acquire central fixation (as in Anderson & Kim, 2019). This was completed to ensure that eye position was calibrated, and that each trial started with gaze located at the center of the screen.

Once fixation was acquired, participants were presented a target cue (an image of the target they needed to search for in the subsequent scene) for 1,000 ms, followed by a second fixation cross for 1,000 ms and then by the search display. The search display consisted of a kitchen scene containing four black shapes—one target and three distractors—all of which were located on either the wall, countertop, or floor. Participants were given the task to search for the shape that matched the target cue and report the orientation of the T (left or right) located in the center of the target shape using the keyboard. If the T embedded within the shape was rotated 90°, left participants responded with the “Z” key, and if the shape was rotated 90°, right participants responded with the “/” key. After each response, a feedback screen was presented for 500 ms that said “correct,” if the participant correctly responded to the orientation of the T; “incorrect,” if the participant incorrectly responded to the orientation of the T; or “too slow,” if more than 15 s elapsed without a response. An illustration of the task can be found in Figure 1A. Eye position was monitored throughout the task. Participants first completed one short training block, and then completed six test blocks.

Data Analysis

We measured RTs, accuracy, and eye position. Eye position analysis was conducted by measuring which shape in each scene was fixated first. Fixations were determined, if eye position fell within a 3.1° × 3.1° region surrounding a shape for 50 ms or longer. Practice trials were excluded from analysis. RTs less than 200 ms and three standard deviations above or below the mean of each condition for each participant were also excluded from analysis.

Only accurate trials were included in RT and first fixation analysis. All dependent variables were analyzed using a 2 (Location Type: High-Probability, Low-Probability) × 2 (Critical Distractor Presence: Present, Absent) repeated-measures analysis of variance (ANOVA). A follow-up analysis was performed in which the selection of trials was restricted such that high- and low-probability trials contained the target in the same spatial location (e.g., the same location on the screen corresponded to a countertop in one scene and the wall in another, which in one case was the high-probability location for a target and in another case was a low-probability location). Spatial overlap was determined if any part of the area of interest that encompassed a shape overlapped with the area of interest encompassing another shape in a different category.

Transparency and Openness

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study, and the study follows The Journal Article Reporting Standards for quantitative research (Appelbaum et al., 2018). All data and all experiment code for all experiments are available on the Open Science Framework (Frandsen & Anderson, 2025). Data were analyzed using R, Version 4.3.2 (R Core Team, 2023), with the packages tidyverse (Wickham et al., 2019) and effectsize (Ben-Shachar et al., 2020). This study’s design and analysis were not preregistered. Due to licensing restrictions, any scene images licensed through Adobe Stock Images cannot be shared; however, a comprehensive list of images is provided.

Results

RT

As shown in Figure 1B, a 2 (Location Type: High-Probability, Low-Probability) × 2 (Critical Distractor Presence: Present, Absent) repeated-measures ANOVA was conducted on RT. There was a significant main effect of location type, $F(1, 39) = 31.72, p < .001, \eta_p^2 = .45$. It was found that trials where the target appeared in its high-probability location ($M = 1,162$ ms, $SD = 133$ ms) were faster than trials where the target appeared in a low-probability location ($M = 1,229$ ms, $SD = 163$ ms). This suggests that search was facilitated when the target appeared in a high-probability location. A significant main effect of critical distractor presence was also observed, $F(1, 39) = 5.77, p = .021, \eta_p^2 = .13$. It was found that participants were slowed when the critical distractor was present ($M = 1,206$ ms, $SD = 156$ ms), relative to when it was absent ($M = 1,185$ ms, $SD = 147$ ms). This suggests that when a possible target is present as a distractor, it interferes with target selection. No interaction was found between location type and critical distractor presence, $F(1, 39) = .38, p = .54, \eta_p^2 = .01$.

Accuracy

The same 2 (Location Type: High-Probability, Low-Probability) × 2 (Critical Distractor Presence: Present, Absent) repeated-measures ANOVA was conducted on accuracy. A significant main effect of critical distractor presence was found, $F(1, 39) = 12.74, p < .001, \eta_p^2 = .25$, indicating that participants were less accurate when the distractor was present ($M = 96.2\%$, $SD = 4.1\%$) than when it was absent ($M = 97.6\%$, $SD = 2.8\%$). No difference in accuracy was found when targets appeared in their high-probability location

($M = 97\%$, $SD = 3.3\%$) compared to their low-probability locations ($M = 96.7\%$, $SD = 3.8\%$), $F(1, 39) = .44$, $p = .51$, $\eta_p^2 = .01$. Nor was an interaction between location type and critical distractor presence observed, $F(1, 39) = .18$, $p = .67$, $\eta_p^2 < .01$.

Eyetracking

Similar to the behavioral results, a 2 (Location Type: High-Probability, Low-Probability) $\times$ 2 (Critical Distractor Presence: Present, Absent) repeated-measures ANOVA was conducted on the probability of fixating the target on the first fixation of a shape (Figure 1C). A main effect of location type was observed, $F(1, 39) = 25.89$, $p < .001$, $\eta_p^2 = .40$, indicating that participants were more likely to fixate the target when it appeared in its high-probability location ($M = 50.4\%$, $SD = 6.2\%$), relative to when it appeared in a low-probability location ($M = 43.2\%$, $SD = 10.3\%$). No reliable difference was found between when a critical distractor was present ($M = 45.9\%$, $SD = 8.9\%$) and when it was absent ($M = 47.7\%$, $SD = 9.5\%$), $F(1, 39) = 4.06$, $p = .051$, $\eta_p^2 = .09$, although the difference was marginally significant and in the same direction as RT. No interaction was found between location type and critical distractor presence, $F(1, 39) = .22$, $p = .64$, $\eta_p^2 = .01$.

A 2×2 repeated-measures ANOVA with the same factors as the previous analysis was conducted on the mean ordinal fixation number (whether the target was the first, second, third, or fourth shape fixated). A main effect of location type was observed, $F(1, 39) = 18.48$, $p < .001$, $\eta_p^2 = .32$, indicating that participants fixated the target on an earlier fixation when it appeared in its high-probability location ($M = 1.69$, $SD = .13$), relative to when it appeared in a low-probability location ($M = 1.82$, $SD = .23$). It was found that participants fixated the target on an earlier fixation when the additional target distractor was absent ($M = 1.74$, $SD = .20$) relative to when it was present ($M = 1.78$, $SD = .20$), $F(1, 39) = 4.56$, $p = .039$, $\eta_p^2 = .1$. No interaction between additional target distractor presence and location type was found, $F(1, 39) = 2.99$, $p = .09$, $\eta_p^2 = .07$.

Supplemental Analyses

To account for possible effects created through relationships between locations within the context of scenes and absolute position in space (e.g., objects located on the wall tend to be higher in the scene), we ran the same 2×2 ANOVA on mean RT but only for trials in which the spatial location of objects was balanced across high- and low-probability trials. There was a significant main effect of location type, $F(1, 39) = 22.84$, $p < .001$, $\eta_p^2 = .37$. It was found that trials where the target appeared in its high-probability location ($M = 1,160$ ms, $SD = 137$ ms) were faster than trials where the target appeared in a low-probability location ($M = 1,217$ ms, $SD = 168$ ms). No reliable difference between when the critical distractor was present ($M = 1,199$ ms, $SD = 162$ ms) and absent ($M = 1,178$ ms, $SD = 149$ ms) was found, $F(1, 39) = 3.55$, $p = .067$, $\eta_p^2 = .08$. No interaction between location type and critical distractor presence was found, $F(1, 39) = .29$, $p = .60$, $\eta_p^2 = .01$.

The same 2×2 repeated-measures ANOVA was conducted on the proportion of first fixation. There was a significant main effect of location type, $F(1, 39) = 16.93$, $p < .001$, $\eta_p^2 = .30$. It was found that the target was more likely to be initially fixated on trials where the target appeared in its high-probability location ($M = 51.2\%$, $SD = 7.1\%$) compared to a low-probability location ($M = 44.9\%$,

$SD = 11.5\%$). There was no reliable difference between when the critical distractor was present ($M = 47.3\%$, $SD = 9.6\%$) relative to when it was absent ($M = 48.8\%$, $SD = 10.4\%$), $F(1, 39) = 1.84$, $p = .183$, $\eta_p^2 = .05$, nor was an interaction between location type and critical distractor presence observed, $F(1, 39) = .33$, $p = .57$, $\eta_p^2 = .01$.

RT and first fixation probability were also analyzed while restricting trials to the first presentation of each scene, to verify that the observed effects were due to the learning of object–location associations and not memory for specific scene–object repetitions. For RT, we found that the effect of location probability remained, $F(1, 39) = 13.87$, $p < .001$, $\eta_p^2 = .26$, the effect of the critical distractor become nonsignificant, $F(1, 39) = 0.21$, $p = .65$, $\eta_p^2 = .01$, with no interaction between them, $F(1, 39) = 1.53$, $p = .22$, $\eta_p^2 = .04$. For first fixation probability, we again found that the effect of location probability remained, $F(1, 39) = 6.97$, $p = .012$, $\eta_p^2 = .15$, and the effect of the critical distractor was nonsignificant, $F(1, 39) = .01$, $p = .92$, $\eta_p^2 = .05$, with no evidence for an interaction between them, $F(1, 39) = 1.53$, $p = .22$, $\eta_p^2 < .01$.

To control for short-term target repetition priming effects on eye movements, we removed trials where the same target appeared in the same location that it appeared in the last time it was presented or the presentation before that and reran the same analysis on RT and first fixation probability. Consistent with the main analysis, for RT location probability remained significant, $F(1, 39) = 16.00$, $p < .001$, $\eta_p^2 = .29$, although there was no significant effect of critical distractor presence, $F(1, 39) = 1.70$, $p = .20$, $\eta_p^2 = .04$, nor a significant interaction, $F(1, 39) = 1.15$, $p = .29$, $\eta_p^2 = .03$. Similarly, for first fixation probability, location probability remained significant, $F(1, 39) = 6.53$, $p = .015$, $\eta_p^2 = .14$, although there was no significant effect of critical distractor presence, $F(1, 39) = 2.62$, $p = .11$, $\eta_p^2 = .06$, nor a significant interaction, $F(1, 39) = .25$, $p = .62$, $\eta_p^2 < .01$.

Discussion

The results for Experiment 1 are consistent with the notion that object–location associations can be formed in the absence of semantic relationships linking them. We found that participants were faster to find target shapes when they appeared in high-probability locations relative to when they appeared in low-probability locations within scenes. This was evident in both our RT and eye movement data. This suggests that repeated exposure to object–location contingencies facilitated more efficient search for objects in naturalistic scenes. This facilitation remained when controlling for possible scene structure confounds like vertical position, indicating that the observed effects are not simply due to the learning of relationships between objects and locations without respect to scene context but instead reflect the learning of statistical regularities between objects and the context within which they are presented. Control analyses ruled out scene-specific memories for object–location pairings and repetition priming of a specific object in a specific location.

We found that the presence of a critical distractor impaired performance, both slowing RTs and reducing accuracy. This shows that when distractors share features that were previously searched for, search performance suffers, likely due to a selection history effect (Anderson et al., 2021). The presence of a critical distractor did not interact with the influence of statistical learning on

attentional guidance, however, suggesting that despite the increased difficulty and the fact that the actual target was no longer the only familiar candidate target shape, participants were still able to robustly leverage the statistical regularities of the targets across scenes.

These results show that scene grammar can be acquired not only through semantic knowledge but also rapidly through statistical regularities within the environment. This suggests a flexible mechanism by which the visual system can build expectations concerning the location of specific objects within a scene that guide attention, even when object–location associations are arbitrary.

Experiment 2

In Experiment 1, we demonstrated that statistical learning can be used to guide attention in naturalistic scenes, but we only utilized a single scene context. In Experiment 2, we tested whether such learning could proceed in a context-dependent manner, such that the context provided by the scene was needed to extract the object–location association of a given shape. We did this by including two additional categories of scene (for a total of three) and varying the association that each target object had across each scene category (e.g., a target object could frequently appear on the countertop in one category of scene, on the floor in another, and on the wall in a third category of scene). If participants were to evidence context-dependent attentional guidance, it would suggest that the attentional system first encodes context before encoding statistical regularities.

Method

Participants

A priori power analysis was conducted using G*Power 3.1 (Faul et al., 2007) for a paired-samples *t* test comparing first fixation proportion between high- and low-probability target locations. Based on the effect size observed in the eyetracking analysis of Experiment 1 ($\eta_p^2 = .40$), we calculated effect size as $d_z = 1.15$. Utilizing this effect size from the previous experiment, the power analysis indicated that a minimum of 10 participants would be sufficient to detect the effect of location type with 80% power at an α level of .05. However, to account for variability in measurement, improve the precision of effect size estimates, and help provide sufficient power to detect a potentially weaker context-dependent effect, we recruited 31 participants from Texas A&M University. Participants received course credit as compensation. The sample included six males, 23 females, and two individuals who chose not to self-report gender. Participants had an average age of 18.59 ($SD = .78$) and were all between the ages of 18 and 21. All participants reported normal or corrected-to-normal vision and normal color vision. All participants provided written informed consent, and all study procedures were approved by the Texas A&M University Institutional Review Board.

Apparatus

Identical to Experiment 1.

Stimuli

Scene stimuli consisted of 40 kitchen, 40 bathroom, and 40 living room scenes in the main blocks and a different set of 12 kitchen,

12 bathroom, and 12 living room scenes in the practice block. The kitchen scenes were selected as a subset of scenes from Experiment 1, and the bathroom and living room scenes were selected through Internet search either from open stock image websites or licensed from Adobe Stock Images. Shape stimuli were identical to the first experiment. For each participant, three shapes from this set were randomly selected to serve as the targets, while the remaining shapes were used as distractors. On each trial, one target and two distractors were placed in each scene. Each target shape was given a different association with either the wall, countertop (raised surfaces in the scene, e.g., a coffee table or a side table in the living room), or floor of the scenes in each scene type (i.e., a target could be associated with the wall in the kitchen, floor in the living room, and the countertop in the bathroom).

Design

The experiment consisted of one practice block of 36 trials and six test blocks of 60 trials for a total of 396 trials (including practice). The practice block consisted of 36 scenes each presented a single time for 36 trials (four trials for each location scene type contingency), where the target appeared in its associated high-probability location (based on its association within each scene) on 100% of trials. The test blocks consisted of 120 scenes that were repeated three times across the task. During the test blocks, targets appeared in their associated location on 75% of trials and across the other two nonassociated locations on the remaining 25% of trials (equally often across the two low-probability locations for each scene type). Each scene type was presented equally often across all trials. On half of all trials, the target appeared with its embedded T rotated 90° left, and the other half of trials, it appeared with its embedded T rotated 90° right. The rotation of the T embedded within the two distractors was randomly determined on each trial with the constraint that all three Ts were not rotated in the same direction. Each target was cued equally often for each scene type. Trial order was randomized across the experiment with the constraint that same scene could not appear within the same block and no more than two consecutive trials could involve the same target in a low-probability location.

Procedure

The procedure was identical to Experiment 1 with the exception that three different scene types were used, and three shapes were placed across the three possible positions within each scene.

Data Analysis

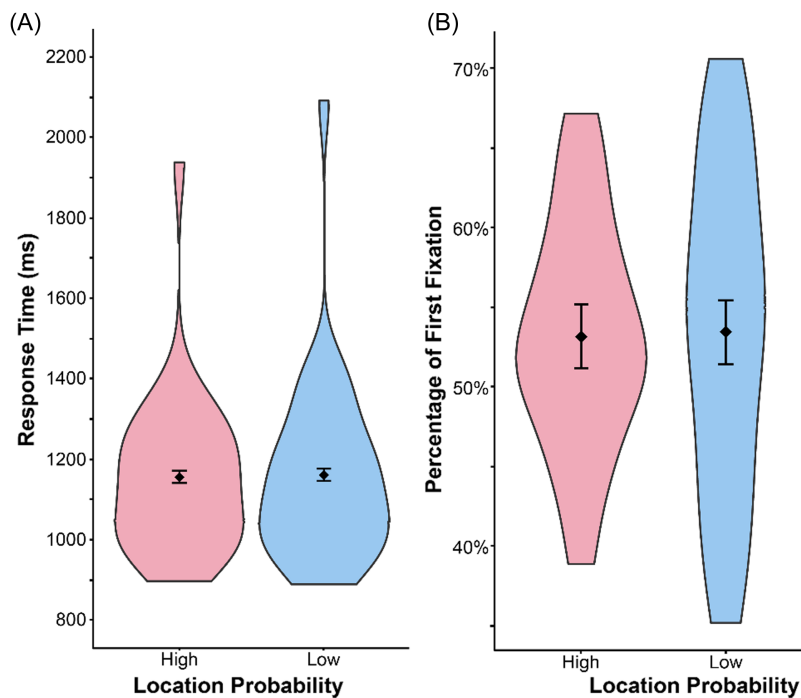
As in Experiment 1, we measured RTs, accuracy, and eye position using the same analytic approach. Because we did not utilize a critical distractor in this experiment, all dependent variables were analyzed using a paired-samples *t* test comparing mean differences for targets appearing in the high-probability location versus a low-probability location.

Results

RT

As shown in Figure 2A, a paired-samples *t* test was conducted to compare high-probability and low-probability locations on RT, which revealed no significant difference between high-probability

Figure 2
Behavioral Results for Experiment 2



Note. (A) Response time across location type in Experiment 2. Violin plots show the distribution of fixation proportions in each condition. Dots represent condition means, and error bars denote 95% Cousineau–Morey confidence intervals. (B) Percentage of first fixation on the target across location type in Experiment 2, shown with the same plotting conventions as in (A). See the online article for the color version of this figure.

locations ($M = 1,156$, $SD = 199$) and low-probability locations ($M = 1,161$, $SD = 230$), $t(30) = -0.50$, $p = .62$.

Accuracy

A paired-samples t test was conducted to compare high-probability and low-probability locations on accuracy, which revealed no significant difference between high-probability locations ($M = 97.9\%$, $SD = 1.9\%$) and low-probability locations ($M = 97.3\%$, $SD = 3.2\%$), $t(30) = 1.44$, $p = .16$.

Eyetracking

As shown in Figure 2B, a paired-samples t test was conducted to compare high-probability and low-probability locations on first fixation probability, which revealed no significant difference between high-probability locations ($M = 53.1\%$, $SD = 7.2\%$) and low-probability locations ($M = 53.5\%$, $SD = 10.1\%$), $t(30) = -0.55$, $p = .58$. Similarly, no significant difference between high-probability ($M = 1.56$, $SD = .12$) and low-probability ($M = 1.54$, $SD = .16$) locations was observed for ordinal fixation count, $t(30) = 1.02$, $p = .32$.

Discussion

In this experiment, we found that participants were unable to learn object–location associations, when these associations

were contingent upon scene context. When the same object was associated with one location within one category of scene and a different location within a different category of scene, no benefit was observed when the target appeared in its associated location compared to a contextually low-probability location. There are multiple potential explanations for the observed lack of contextually dependent learning. One possible explanation is that participants were not sufficiently exposed to each object–scene association to learn the relationship. Alternatively, it may be that object–location associations are formed in a manner that is insensitive to scene context and therefore generalizes across categories of scene, which we tested in Experiment 3.

Experiment 3

In Experiment 3, to test if object–location associations are learned across contexts, we utilized a design with two scene types—one with object–location associations and one without—and presented them intermixed. If participants demonstrate prioritization of a high-probability location within scenes for which that association is and is not reflective of the underlying object–location statistics, it would suggest that the learning of object–locations associations that guide attention generalizes across, or is otherwise insensitive to, the context provided by the scene.

Method

Participants

A priori power analysis was conducted using G*Power 3.1 (Faul et al., 2007) for a 2×2 repeated-measures ANOVA. Based on the effect size observed in Experiment 1 ($\eta_p^2 = .40$), we calculated the effect size as $f = .81$. Utilizing this effect size to calculate the effect size of the experiment and assuming a moderate correlation among repeated measures ($\rho = .5$), the analysis indicated that a minimum of four participants would be sufficient to detect at least one simple effect with 80% power ($\alpha = .05$). However, to account for variability in measurement, improve the precision of effect size estimates, to help provide sufficient power to detect a potentially weaker effect due to the introduction of a nonpredictive context, and to account for possible interaction effects, we recruited 45 participants from Texas A&M University. The sample included 15 males, 31 females, and two individuals who chose not to self-report gender. Participants had an average age of 19.17 ($SD = 1.23$) and were all between the ages of 18 and 24. All participants reported normal or corrected-to-normal vision and normal color vision. All participants provided written informed consent, and all study procedures were approved by the Texas A&M University Institutional Review Board.

Apparatus

Identical to Experiment 1.

Stimuli

Scene stimuli consisted of 64 kitchen and 64 bathroom scenes in the main blocks and a different nine kitchen and nine bathroom scenes in the practice block. The kitchen scenes were selected as a subset of scenes from Experiment 1 and the bathroom scenes from Experiment 2, in addition to scenes that were selected via Internet search either from open stock image websites or licensed from Adobe Stock Images. Shape stimuli were identical to Experiment 1. Target and distractor shapes were determined as in Experiment 2. Each target shape was given a different association with either the wall, countertop, or floor of the scenes in one scene type. The scene type assigned to have predictive object–location associations (either the kitchen or bathroom) was counterbalanced across participants.

Design

The design was identical to Experiment 2 with the following exceptions. The experiment consisted of a practice block (18 trials) and six test blocks (64 trials each) for a total of 402 trials (including practice). In all blocks, half of the trials consisted of the scene type with predictive object–location associations, and the other half consisted of scenes that lacked object–location associations (i.e., objects were equally likely to appear across the three location types). During practice, on the trials where the targets had an association within that scene context, targets appeared in the associated location on all trials. In the context without the associated location, targets appeared in each location type equally often. Scenes used in practice were only presented one time during the practice and not presented again during the task.

The test blocks consisted of 128 scenes that were repeated three times across the task. During the test blocks, on trials where

predictive scene type was presented, targets appeared in their associated location on 75% of trials and across the other two nonassociated locations on the remaining 25% of trials (each equally often). In the unpredictable scene type, targets appeared in each of the three locations equally often.

Procedure

The procedure was identical to Experiment 2 with the exception that two different scene types were used.

Data Analysis

As in Experiments 1 and 2, we measured RTs, accuracy, and eye position using the same analytic approach. All dependent variables were analyzed using a 2 (Location Type: High-Probability, Low-Probability) $\times$ 2 (Scene Type: Predictive, Unpredictive) repeated-measures ANOVA. As the unpredictable category of scene does not have location associations, the location factor at this level of scene type is with reference to the probabilities in effect for the other category of scene.

Results

RT

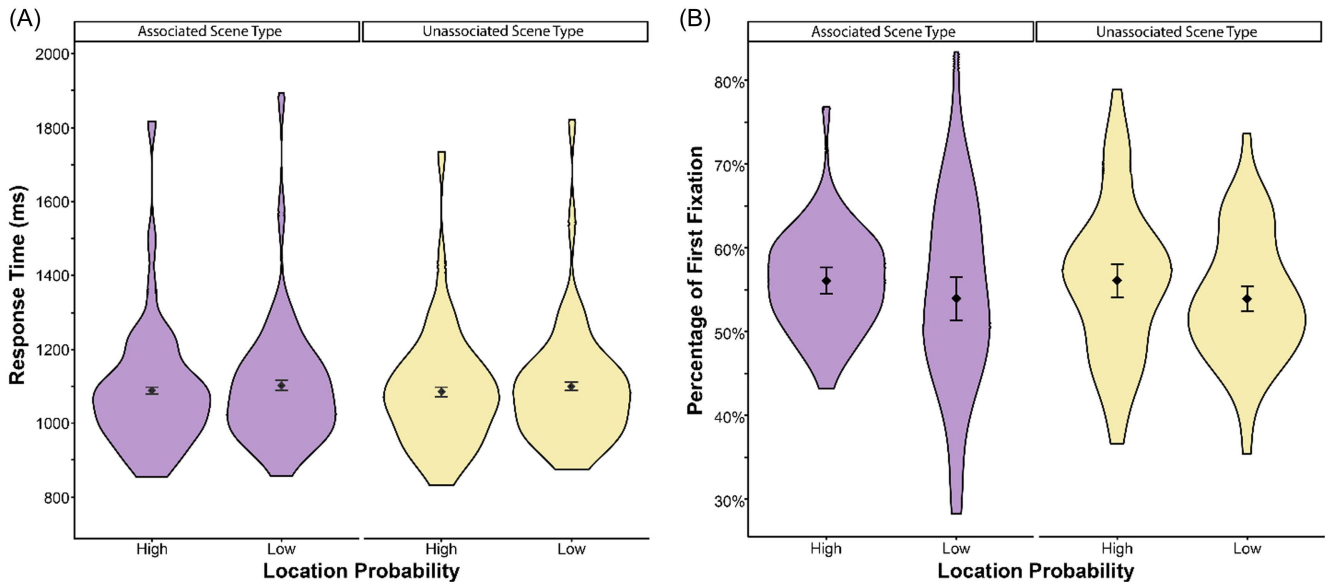
As shown in Figure 3A, a 2 (Location Type: High-Probability, Low-Probability) $\times$ 2 (Scene Type: Predictive, Unpredictive) repeated-measures ANOVA was conducted on RT. There was a significant main effect of location type, $F(1, 44) = 4.604, p = .038, \eta_p^2 = .09$. It was found that trials where the target appeared in its high-probability location ($M = 1,087$ ms, $SD = 165$ ms) were faster than trials where the target appeared in a low-probability location ($M = 1,101$ ms, $SD = 25.6$ ms). No difference between the predictive ($M = 1,095$ ms, $SD = 173$ ms) and unpredictable ($M = 1,093$ ms, $SD = 24.4$ ms) scene types was observed, $F(1, 44) = .13, p = .72, \eta_p^2 < .01$, nor was the interaction between location type and scene type significant, $F(1, 44) = .01, p = .94, \eta_p^2 < .01$.

Accuracy

A 2 (Location Type: High-Probability, Low-Probability) $\times$ 2 (Scene Type: Predictive, Unpredictive) repeated-measures ANOVA was conducted on accuracy. No difference between high-probability ($M = 98.3\%$, $SD = 1.94\%$) and low-probability ($M = 98.2\%$, $SD = 2.47\%$) probability locations was found, $F(1, 44) = .26, p = .61, \eta_p^2 < .01$. No difference between predictive ($M = 98.1\%$, $SD = 2.19\%$) and unpredictable ($M = 98.4\%$, $SD = 2.25\%$) scenes was found, $F(1, 44) = 1.97, p = .17, \eta_p^2 = .04$, nor was there an interaction between location type and scene type, $F(1, 44) = .02, p = .90, \eta_p^2 < .01$.

Eyetracking

As shown in Figure 3B, a 2 (Location Type: High-Probability, Low-Probability) $\times$ 2 (Scene Type: Predictive, Unpredictive) repeated-measures ANOVA was conducted on the probability of fixating the target on the first fixation of a shape. No reliable difference between high-probability ($M = 56.1\%$, $SD = 7.74\%$) and low-probability ($M = 53.9\%$, $SD = 9.65\%$) locations was observed,

Figure 3*Behavioral Results for Experiment 3*

Note. (A) Response time across location type and scene association in Experiment 3. Violin plots show the distribution of fixation proportions in each condition. Dots represent condition means, and error bars denote 95% Cousineau–Morey confidence intervals. (B) Percentage of first fixation on the target across location type and scene association in Experiment 3, shown with the same plotting conventions as in (A). See the online article for the color version of this figure.

$F(1, 44) = 3.92, p = .054, \eta_p^2 = .08$, although the difference was marginally significant and in the same direction as RT. No difference was found between predictive ($M = 55.0\%$, $SD = 9.14\%$) and unpredictable ($M = 55.0\%$, $SD = 8.47\%$) scene types, $F(1, 44) = 0, p = .99, \eta_p^2 < .01$, nor was there an interaction between location type and scene type, $F(1, 44) = 0, p = .95, \eta_p^2 < .01$.

A 2×2 repeated-measures ANOVA with the same levels as the previous analysis was conducted on the mean ordinal fixation number. We found no difference between high-probability ($M = 1.51$, $SD = .12$) and low-probability ($M = 1.54$, $SD = .14$) locations, $F(1, 44) = 2.87, p = .097, \eta_p^2 = .06$. No difference between predictive ($M = 1.52$, $SD = .13$) and unpredictable ($M = 1.52$, $SD = .13$) scene types was found, $F(1, 44) = .04, p = .85, \eta_p^2 < .01$, nor was there an interaction between location type and scene type, $F(1, 44) = .74, p = .39, \eta_p^2 = .02$.

Discussion

The RT data indicate that targets appearing a high-probability location were identified more quickly regardless of whether they were embedded in a predictive or unpredictable context. Although this effect is small and only modestly significant, we cautiously suggest that participants develop global region-level biases for each object across scenes. By this account, observers do not consider the context provided by the scene beyond how the scene is segmented, collapsing their learning across kitchens and bathrooms. Coupled with the results of Experiment 2, this may be due to a lack of attention to the context provided by the category of scene during learning or potentially due to an assumption that

the associated location of an object should be stable across categories of scenes. An effect of object–location probabilities was not observed in accuracy or eyetracking analyses, although in the case of the latter was marginally significant. The overall reduction in effect size across measures compared to Experiment 1 may be explained, at least in part, by the fact that if learning is not sensitive to the context provided by the category of scene, the object–locations associations are overall less predictive (54.17% predictive across all trials) as compared to Experiment 1 (75% predictive).

General Discussion

The current set of experiments demonstrates that statistical learning of object–location associations can guide attention during visual search in naturalistic scenes, even when the objects themselves lack semantic relationships with the content of the scenes. In Experiment 1, we associated novel objects (lacking semantic meaning) with a given region of a scene by manipulating the probability, it would occur across regions of the scene (e.g., wall, countertop, floor). We found that participants successfully learned associations between abstract shapes and specific regions of a scene and that this learning facilitated search performance, indicating that attentional guidance to objects in real-world scenes can emerge purely from statistical regularities.

In Experiment 2, we varied the association that each target shape had with locations within scenes across three different categories of scene. The results revealed that when object–location associations varied according to the context provided by the category of scene,

participants failed to show evidence of location-based guidance. This suggests a limitation in the attentional system's ability to form context-dependent object–location associations via statistical learning, when a specific association must be activated by the scene context. Further, Experiment 3 showed that object–location associations learned in a predictive context are similarly applied in an unpredictable context. This suggests that the visual system may form inflexible attentional templates that are sensitive to the layout of scenes but are applied similarly across contexts.

Experiment 2 found that object–location associations could not be learned when these associations varied across different scene contexts. This could be the result of a variety of factors. One possible explanation was that there were too many contingencies to keep track of and update with experience, making it impractical to learn the object–location associations. More robust learning might have become evident with more protracted training, and learning might have been facilitated with greater exposure to the scene context prior to engaging in search (e.g., previewing the scene). Additionally, the learning of object–location associations may be limited in its sensitivity to scene context, which is consistent with the results of Experiment 3 showing a similar consequence of statistical learning in predictive and unpredictable contexts.

Together, these findings present new evidence for how attentional templates for objects are formed and deployed in visual search. Many previous studies have shown that semantic relationships between objects are able to be leveraged by the attentional system to guide search (e.g., [Henderson & Hayes, 2017](#); [Vö & Wolfe, 2013](#)); however, it is unclear how these templates emerge due to their usage of real-world objects with existing associations. In the present study, by using novel objects (lacking existing semantic associations) and associating them with specific regions of a scene, we show that statistical learning is sufficient to guide attention to object-associated regions during visual search.

Existing theories suggest that statistical learning and semantic knowledge are supported by distinct memory systems ([Hutchinson & Turk-Browne, 2012](#)). Semantic knowledge is believed to be an explicit form of memory, whereas statistical learning is thought to be implicit. Even though they are supported by separate systems, they have similar effects on the guidance of attention, potentially modulating attentional priority according to similar and overlapping principles reflected in models of selection history ([Anderson et al., 2021](#); [Awh et al., 2012](#)). When meaningful objects are presented with spatial regularities in a manner consistent with semantic knowledge concerning these objects, an effect of semantics and statistical learning may separately guide attention in parallel or semantic knowledge may come to dominate attentional guidance through a unified attentional template, with statistical learning serving as more of a precursor to semantic guidance. Likewise, the relationship between statistical learning and the formation of the kind of semantic knowledge that guides attention is unclear. Further work needs to be done to investigate whether statistical learning contributes to the development of the semantic knowledge that guides attention in scenes and if attentional guidance by semantic knowledge can emerge in the absence of statistical learning.

Attention is automatically captured by objects that are semantically inconsistent with the environment ([Henderson et al., 1999](#); [Lofus & Mackworth, 1978](#); [Vö & Henderson, 2009](#)). Such semantic

guidance often depends on the broader meaning of the scene and is, by definition, caused by a learning-dependent influence that does not generalize across contexts. In contrast, statistical learning may operate at a different representational level, binding objects to spatial or structural properties of scenes, allowing attentional guidance to function across contexts that share similar spatial layouts even when semantic consistency is absent ([Brockmole & Henderson, 2006](#); [Chun & Jiang, 1998](#); [Y. Jiang & Wagner, 2004](#)). Contextual cueing paradigms find that when repeatedly exposed to the same layout, global scene structure can be utilized to guide search ([Brockmole et al., 2006](#); [Chun & Jiang, 1998](#)). Our findings also showcase that, similar to contextual cueing, repeated exposure to target objects embedded within a predictable layout can be leveraged to guide attention; however, unlike contextual cueing, which has limited tolerance to variation across predictive spatial configurations, we find that statistical learning can operate on the level of regions within scenes that vary nontrivially across individual scenes in their specific configuration, suggesting more flexible learning and generalization to novel scene layouts.

Statistical learning can support the formation of a variety of associations that can subsequently guide attention during visual search. For example, [Bahle et al. \(2021\)](#) showed that repeated co-occurrence of a visual feature (e.g., color) with a category of object can bias attention toward stimuli rendered in the associated color during search for a category member (e.g., exposure to red cars will facilitate search for a car when the target car is red). [Kershner and Hollingworth \(2022\)](#) extended this principle to search within naturalistic scenes, demonstrating that observers can learn associations between object features and scene contexts that subsequently guide search behavior in a context-dependent manner. Similar to the way learning is deployed in this study, [Clement and Anderson \(2024\)](#) demonstrated that statistical learning can apply to not only the features of a searched-for object but also to the relationship between those features and the features of a neighboring object encountered during search. These studies showcase that statistical learning can operate over multiple representational levels (general visual features, specific object identities, or spatial structures of environments) and that each level can influence attentional guidance. Our findings extend this framework by demonstrating that object–location associations can be established even when the objects lack semantic relevance to the location, and that such associations create region-level biases across contexts that share similar spatial layouts. Here we demonstrate that, through statistical learning, features of the environment can be bound with individual objects that are not semantically related to that environment. This could be one potential mechanism for the creation of an attentional template for an object before higher level semantic understanding has been formed.

In [Brockmole and Le-Hoa Vö \(2010\)](#), participants were tasked to search for a target letter embedded within bedroom scenes where target letters either co-occurred on a pillow (predictable) or in random positions in the room (unpredictable). They found that participants were faster to find the target when it co-occurred with the pillow. They also found that this learning transferred to living room scenes (semantically consistent) but did not transfer to kitchen scenes (semantically inconsistent). Unlike these results, we found that the learning in our study is applied across contexts provided by the category of scene. One likely reason for this is that they

associated the target with a semantically meaningful object (e.g., a pillow), whereas we associated objects with regions of a given scene (i.e., wall, counter, or floor). In our study, the context nonspecific learning may occur, because these regions hold similar functional significance across scene types given the lack of semantic information represented by the searched-for objects. For example, a hand tool such as a screwdriver is likely to be found on a raised surface (e.g., counters or tables), because raised surfaces are generally closer to a person's hands, making it easier to grasp. In our study, the association between an object and a location lacked such functional significance, which may play a prominent role in context-dependent attentional guidance within scenes.

It is difficult to dissociate to what degree a given attentional effect is attributed to semantic knowledge or statistical learning when using real-world objects to study attention. Over an individual's lifespan, they are typically exposed to a given real-world object numerous times, where it has a true probability of occurring across different locations. The use of real-world objects in studies of attention confounds the role of statistical learning with semantic knowledge, because over the lifespan, an individual has already learned the statistical probability with which a given object occurs across locations and also has semantic knowledge about the object going into the study. It becomes difficult to compare the results of previous studies with the current because if you observe stronger guidance in search for real-world objects, it may be stronger because learning in the present study is restricted to the context of the experiment whereas real-world objects tap into learning that has occurred over the lifetime both with respect to semantics and statistics.

Previous evidence suggests that that statistical learning is commonly implicit, and observers may lack awareness of statistically learned relationships that nonetheless guide attention (Chun & Jiang, 1998, 2003; Ferrante et al., 2018; B. Wang & Theeuwes, 2018). However, other evidence suggests that such attentional effects might also be at least partially driven by awareness (Smyth & Shanks, 2008; Vadillo et al., 2016, 2020). We did not test for awareness in this study to minimize the prospect that probing awareness of the manipulation may bring forth awareness of the manipulation, so we cannot speak to whether participants were aware of their own learning of the object–location associations, especially in Experiment 1 where learning was more robust. Regardless of whether participants were aware of their learning of the object–location contingencies, however, attentional guidance was brought about via statistical regularities presented in the experiment that lack semantic associations, suggesting that such guidance can occur even when the associations responsible for it are arbitrary and unrelated to the function of objects.

In conclusion, we found that object–location associations could emerge via statistical regularities to guide attention in the absence of object-related semantic information. Unlike guidance via semantic knowledge, this statistically learned guidance is not context-specific, consistent with other demonstrations of statistical learning effects on attention allocation (e.g., Y. Jiang & Chun, 2001; Zhao et al., 2013). Overall, our findings suggest a specific role for statistical learning in attentional guidance by scene grammar, which confers the ability to rapidly prioritize locations within a scene where an object is more likely to be encountered even if the underlying relationship is unrelated to the function of the object.

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